SUBJECT CODE : 23A01702b

RAILWAYS, AIRPORTS, DOCKS AND HARBOUR ENGINEERING

UNIT – IV : GEOMETRIC DESIGN OF RUNWAYS AND TAXIWA

(Handwritten Study Notes - Exam Oriented)

Syllabus (as given):

Aircraft Characteristics – Influence of Characteristics on Airport Planning and Design – Geometric Design Elements of Runway – Standards and Specifications – Functions of Taxiways – Taxiway Geometric Design – Geometric Elements and Standard Specifications – Runway and Taxiway Lighting

1. Aircraft Characteristics

The design of an airport depends on the characteristics of the aircraft that will use it. The most demanding aircraft is called the critical (design) aircraft.

Characteristic	Meaning / notes
Weight	MTOW (max take-off weight) = OEW (operating empty weight) + payload + fuel. Also MLW (max landing weight), MZFW (max zero-fuel weight)
Size	Wing span, fuselage length, height of tail, wheel base, wheel track (outer main gear wheel span – OMGWS)
Landing gear	Single, dual, dual-tandem, double dual tandem (wide bodies); decides load distribution & tyre pressure
Speed & range	Approach speed, take-off speed, cruise speed; range with full payload
Capacity	Number of passengers / cargo payload (e.g. A320 ≈ 150-180, B777 ≈ 300-400, A380 ≈ 500+)
Turning radius	Minimum radius of turn on ground depends on wheel base and nose-wheel steering angle

UNIT – IV : Geometric Design of Runways and Taxiways

Characteristic	Meaning / notes
Engine type	Piston, turbo-prop, turbo-jet, turbo-fan; decides noise, jet blast and take-off length
Fuel, noise, emissions	Affect environment, land use zones around the airport

Aircraft weight categories (wake turbulence – ICAO)

- Light (L) ≤ 7000 kg; Medium (M) 7000 – 136 000 kg; Heavy (H) > 136 000 kg; and Super (A380). Used for runway separation.

Examples of wing span and ICAO code

Aircraft	Wing span	Code letter
ATR 72 (turboprop)	≈ 27 m	C
A320 / B737	≈ 34 – 36 m	C
B777 / A350	≈ 61 – 65 m	E
A380	≈ 80 m	F

2. Influence of Aircraft Characteristics on Planning and Design

Characteristic	Influence on airport
Weight (MTOW)	Determines pavement thickness, strength (ACN-PCN), runway length, taxiway and apron strength
Wing span	Fixes width of runway, taxiway, apron, separation between taxiways, gate spacing, hangar width
Fuselage length	Decides size of apron gate, hangar length, parking position and turning area
Wheel base & wheel track	Decide turning radius, taxiway width and fillet radius, the pavement edges
Tail height	Decides OLS, height of hangar doors, clearance near approach
Tyre pressure & gear type	Type of pavement (rigid for high pressure), pavement design and ACN

Characteristic	Influence on airport
Speed (landing & take-off)	Runway length, exit taxiway geometry (rapid exit), sight distance, vertical curves
Passenger capacity	Terminal building area, number of gates, parking and road system
Fuel capacity / range	Runway length for long-haul flights, fuel farm and hydrants size
Noise & jet blast	Land-use zoning, blast fences, location of terminal and taxiways

- In short: weight → strength, size → width/clearance, speed → length/curves, capacity → terminal.

3. Geometric Design Elements of Runway

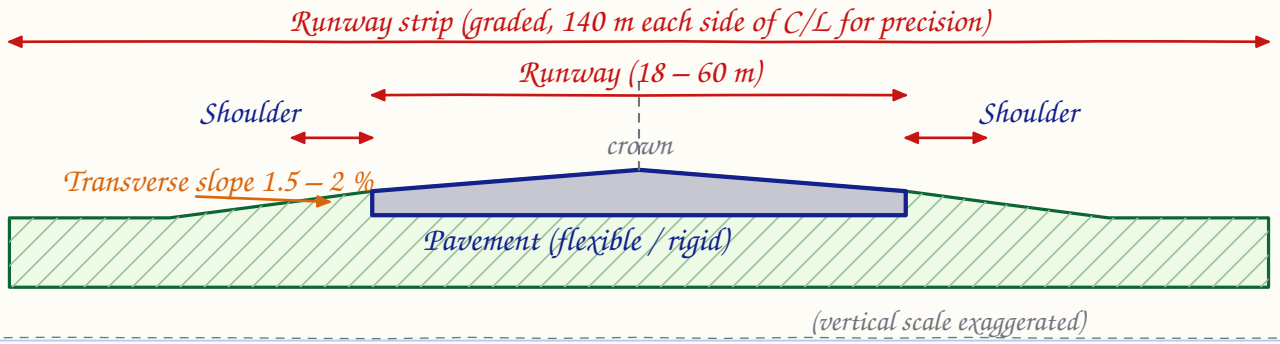


Fig: Cross-section of a runway and strip

(a) Runway length

Obtained from basic length with corrections for elevation, temperature and gradient (Unit III). Length also depends on the critical aircraft, take-off weight and surface (dry/wet).

(b) Runway width

Code number	Width (m)	Remarks
1	18	small aircraft
2	23	
3	30	45 m for code letter D/E
4	45	60 m for code letter F

- Shoulders on both sides, strong enough for occasional run-off and vehicles; overall width

of runway + shoulders ≥ 60 m for code letters D/E and 75 m for F.

(c) Longitudinal gradient

Item	Code 3 and 4	Code 1 and 2
Max effective gradient	1.0 %	2.0 %
Max longitudinal slope	1.25 % (0.8 % in first and last quarter for Code 4)	2 %
Rate of change of slope	0.1 % per 30 m	0.2 % per 30 m
Vertical curve radius (min)	30 000 m (Code 4); 15 000 m (Code 3)	7 500 m
Line-of-sight	3 m above any point must be visible from any other point 3 m above, within half runway length	2 m (Code 2); 1.5 m (Code 1) up to half runway length

- **Effective gradient** = (max. elevation − min. elevation along centre line) ÷ length × 100.

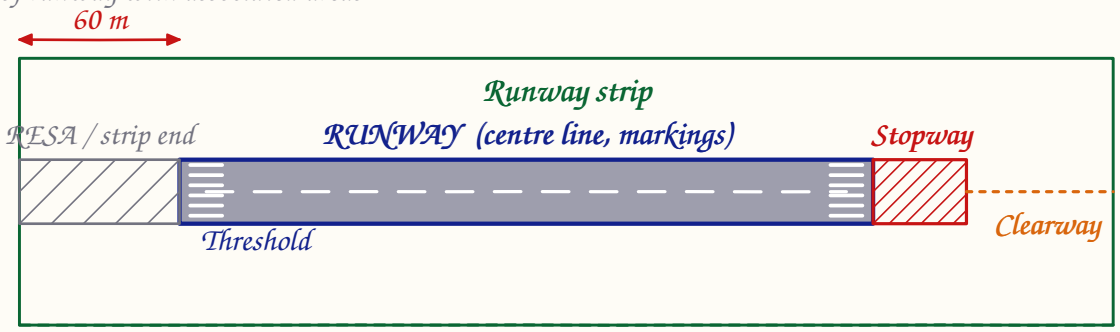
  Example: $L = 2400$ m, difference = 7.2 m → $G = 7.2/2400 \times 100 =$ **0.3 %** (< 1 %, OK).

(d) Transverse gradient

- **1.5 %** for code letters C, D, E, F; **2 %** for A and B (crowned section) to drain water fast; shoulders 2.5 % max. Steeper slopes make aircraft unstable.

(e) Runway strip and safety areas

Plan of runway with associated areas



Strip extends 60 m beyond runway end / stopway; RESA ≥ 90 m (240 m recommended)

Fig: Runway strip, stopway, clearway and RESA

- **Runway strip**: cleared, graded area around the runway to reduce damage to aircraft that run off; width on each side of centre line: **140 m** (precision approach, Code 3/4), **75 m** (non-instrument Code 3/4); extends **60 m** beyond runway end.
- **RESA** (runway end safety area): at least **90 m** (recommended 240 m) beyond strip end,

width ≥ 2 × runway width; graded to stop an aircraft overrunning.

- **Stopway**: same width as runway, strong enough to stop aircraft in an aborted take-off (no landings). **Clearway**: obstacle-free area beyond the runway, width ≥ 150 m, length ≤ half of TORA.

(f) Runway markings

Marking	Description
Designation (number)	Two-digit number = magnetic bearing ÷ 10, e.g. 09, 27 (L/C/R for parallel)
Centre line	Broken white line, 30 m stripe + 20 m gap; gives alignment
Threshold	Series of white longitudinal stripes at the start of landing area
Aiming point	Two broad white bars ~ 300 m from the threshold
Touchdown zone	Pairs of white bars at 150 m interval for precision runways
Edge marking	Continuous white lines on both sides

(g) Pavement

- **Flexible** (bituminous) or **rigid** (concrete, PQC) or composite pavement; designed using **ACN – PCN** method: aircraft is allowed if **ACN ≤ PCN**.

4. Taxiways

Functions of taxiways

- Provide **safe and quick movement** of aircraft between runway, apron, hangar and other areas.
- Allow aircraft to **leave the runway quickly** after landing, so the runway capacity is increased.
- Give access to take-off position, and keep ground movement **separate** from the runway.

Types of taxiways

Type	Description
Parallel (entrance) taxiway	Runs parallel to the runway; used to move aircraft to the runway end; avoids back-tracking

UNIT – IV : Geometric Design of Runways and Taxiways

Type	Description
Exit (turn-off) taxiway	Joins runway at an angle; right-angle exit (90°) or rapid exit / high-speed exit at 30°
Apron taxiway / taxilane	Passes through apron; gives access to aircraft stands
Bypass taxiway	Near the runway end to allow aircraft to pass the waiting ones
Holding bay / holding apron	Area near runway end where aircraft wait for clearance

5. Taxiway Geometric Design

Taxiway turn-offs

V (km/h), f = 0.13; W wheel base, T taxiway width, S = mid main-gear to edge

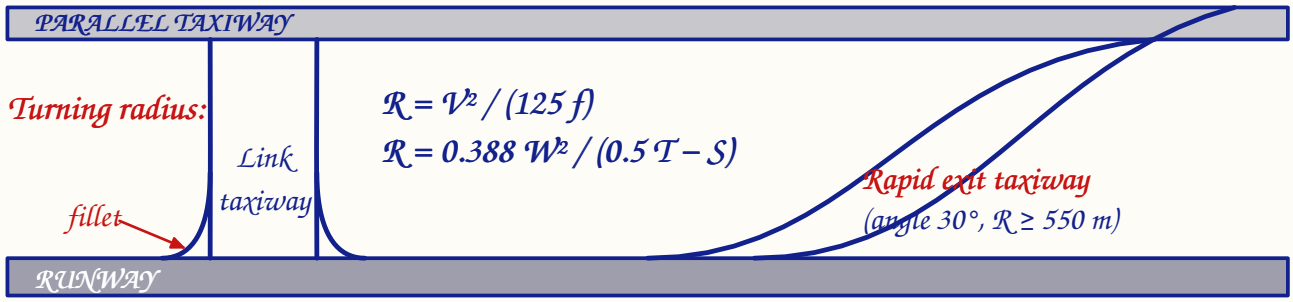


Fig: Taxiway turn-offs and turning radius

Standards (ICAO)

Element	Code letter A, B	Code letter C – F
Longitudinal slope (max)	3 %	1.5 %
Rate of change of slope	1 % per 25 m	1 % per 30 m
Transverse slope	2 %	1.5 %
Sight distance (eye at 1.5 m – 3 m)	visible for 150 m	visible for 300 m (3 m above)
Minimum width of taxiway	A: 7.5 m; B: 10.5 m	C: 15 m (18 m); D: 18 m (23 m); E: 23 m; F: 25 m

- Separation between runway centre line and taxiway centre line (instrument Code 3/4) is about 150 – 180 m, depending on code letter; between taxiways about 1.2 × wing span + clearance.

- **Rapid exit taxiway**: angle **30°**; designed for exit speed **93 km/h** (Code 3/4), turning curve radius ≥ **550 m**, followed by 75 m straight; saves runway occupancy time.
- **Right-angle exit**: 90° with fillets; exit speed ≈ 30 km/h; requires fillet radius.

Radius of turning curve

$$R = V^2 / (125\, f)$$

V = speed of aircraft (km/h), f = coefficient of friction = **0.13** (side friction). Ex: V = 45 km/h → $R = 45^2 / (125 \times 0.13) = 2025 / 16.25 =$ **124.6 ≈ 125 m**.

$$R = 0.388\, W^2 / (0.5\, T - S)$$

W = wheel base (m), T = width of taxiway pavement, S = distance from mid-point of main gear to the edge of pavement. **Ex:** W = 20 m, T = 23 m, S = 5.5 m → $R = 0.388 \times 400 / (11.5 - 5.5) = 155.2 / 6 =$ **25.9 m**. A **fillet** is added at junction to avoid wheels leaving the pavement.

- **Shoulders** on taxiway (stabilised) to prevent erosion by jet blast; overall width of taxiway + shoulders ≈ 25 m (C) to 44 m (E/F).
- Holding positions with **stop bars** are marked before entering the runway.

6. Runway and Taxiway Lighting

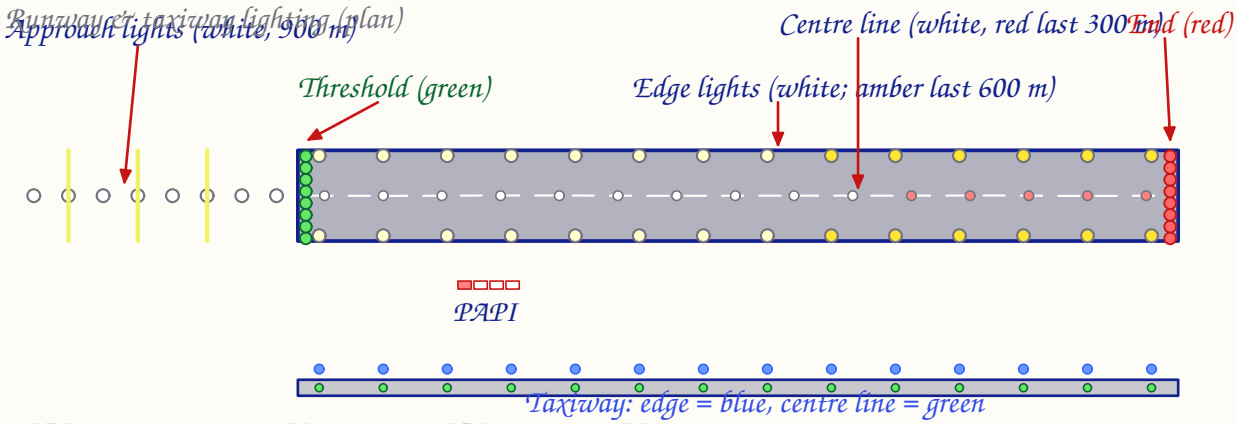


Fig: Airport lighting layout

Light	Colour & arrangement
Approach lighting	Row of white lights along the extended centre line (900 m for CAT I/II/III) with cross-bars (barrettes) – gives direction and distance at night/fog
Threshold lights	**Green** lights across the runway at the threshold (outward-facing)

Light	Colour & arrangement
Runway edge lights	White, along both edges at ≤ 60 m; yellow (amber) for last 600 m / one third (caution zone)
Runway end lights	Red – at the end of runway
Runway centre-line lights	White at 30 m (15 m for CAT III); alternate red-white for next 600 m; red in last 300 m
Touchdown zone lights	White barrettes in the touch-down zone
PAPI / VASI	Visual glide slope: 4 units – white and red lights tell if aircraft is above or below the 3° glide path
Taxiway centre line	Green (for guidance on taxiway); in the exit curves green-yellow for ILS sensitive area
Taxiway edge lights	Blue along edges, at ≤ 60 m
Stop bars / holding	Red across taxiway at holding position; runway guard lights yellow
Aerodrome beacon	Rotating white / green flashing light to identify the airport
Obstruction lights	Red fixed lights on tall objects; apron flood lights for stands

- Lighting requirements: high reliability, intensity control (3 – 5 step), power with secondary source and fast switch-over (e.g. ≤ 1 s for CAT III).
- Taxiway markings are yellow, runway markings are white; instruction signs: mandatory (red/white) and information (yellow/black).

7. Pavement Strength: ACN and PCN

ACN (Aircraft Classification Number) expresses the relative damaging effect of an aircraft on a pavement of a given subgrade strength. PCN (Pavement Classification Number) expresses the bearing strength of the pavement.

- Rule: an aircraft can use the pavement without restriction if ACN ≤ PCN.
- PCN is reported as a code: e.g. PCN 80 / R / B / W / T = number / pavement type (R rigid, F flexible) / subgrade category (A–D) / tyre pressure (W high, X medium, Y low, Z very low) / method (T technical, U using aircraft).

- Subgrade categories: A high, B medium, C low, D ultra-low strength.
- Flexible pavement design for airports: CBR method, FAA method; rigid: Westergaard's stress analysis, PCA method; thickness depends on MTOW, gear type and annual departures.

8. Apron, Holding Bay and Signs

Apron

- Area where aircraft are parked for passenger/cargo loading, fuelling, servicing; adjacent to terminal; strong rigid pavement, slope 0.5 – 1 %.
- Size depends on number of gates, aircraft size (wing span, length) and parking method (nose-in, angled, parallel).
- Wing-tip clearance at stands: 3 m (Code C), 4.5 m (D, E), 7.5 m (F).

Holding bay

- A paved area near the runway end where an aircraft waits for take-off clearance, engine run-up, or to let other aircraft pass; keeps the taxiway free, increases capacity.

Taxiway markings and signs

Item	Description
Taxiway centre-line	Continuous yellow line
Holding position marking	Four lines (two continuous + two dashed) across the taxiway, yellow
Mandatory signs	White on red – runway designation, holding position, no entry
Information signs	Yellow with black letters – taxiway designations, direction, location

9. Worked Examples

Example 1 – Runway pavement width check

Critical aircraft B777 (wing span ≈ 65 m, code 4E). From the table width = 45 m; shoulders 7.5 m each side gives 60 m overall (min. for letter E).

Check: transverse slope = 1.5 % → rise at crown over 22.5 m half-width = 0.015 × 22.5 = 0.34 m.

Example 2 – Rate of change of slope

Two grades meet: +0.6 % and −0.4 % on a Code 4 runway. Change = 1.0 %.

Max rate 0.1 % per 30 m → vertical curve length = 1.0 / 0.1 × 30 = 300 m.

Check with radius: R = 30 000 m per 1 % change → $L = R \times A / 100 = 30\,000 \times 1.0 / 100 =$ 300 m (same).

Example 3 – Taxiway turning radius

Speed on curve 65 km/h, $f = 0.13$: $R = 65^2 / (125 \times 0.13) = 4225 / 16.25 =$ 260 m.

For speed 30 km/h: R = 900 / 16.25 = 55.4 m (right-angle exit).

Example 4 – Longitudinal gradient check

Runway length 2800 m; levels at start = 112.0 m, middle = 114.8 m, end = 113.2 m. Highest − lowest = 114.8 − 112.0 = 2.8 m.

Effective gradient = 2.8 / 2800 × 100 = 0.10 % < 1.0 % → OK.

10. Navigational Aids (Brief)

- ILS (Instrument Landing System): localizer gives lateral guidance, glide path gives vertical guidance (3° slope), marker beacons / DME give distance.
- VOR / DME / NDB radio aids for en-route navigation; GNSS / GAGAN (satellite navigation) for approach with vertical guidance.
- CAT I / II / III precision approaches: decision height ≥ 60 m (CAT I), 30 m (CAT II), lower / zero (CAT III) – approach and centre-line lighting standard increases.
- Windsock / wind direction indicator, signalling area, aerodrome beacon and Runway visual range (RVR) equipment give visual information to the pilot.

11. Summary Tables

Element	Runway	Taxiway
Function	Take-off and landing	Movement between runway, apron, hangars
Longitudinal slope (max)	1 % effective (Code 3/4)	1.5 % (C–F), 3 % (A, B)
Transverse slope	1.5 – 2 %	1.5 – 2 %
Width	18 – 60 m	7.5 – 25 m
Edge light colour	White (amber last 600 m)	Blue
Centre-line light	White / red	Green
Markings	White	Yellow

Remember

- Weight → pavement strength; wing span → widths & separations; wheel base → turning radius.
- Runway edge white, threshold green, end red, taxiway edge blue, taxiway centre green.
- Rapid exit: 30°, exit speed 93 km/h, radius ≥ 550 m.
- ACN ≤ PCN for unrestricted use.

Important Questions – Unit IV

Short answer (2 marks)

- Define critical aircraft, MTOW and wheel base.
- Give two effects of wing span on airport design.
- Write standard transverse slope of runway and ICAO runway widths.
- What is a rapid exit taxiway? Write its angle.
- State the colours of runway edge, threshold, end and taxiway lights.

UNIT – IV : Geometric Design of Runways and Taxiways

Long answer (10 marks)

- Explain aircraft characteristics and their influence on airport planning and design.
- Explain geometric design elements of a runway with ICAO standards.
- Discuss the functions and types of taxiways; explain taxiway geometric design elements.
- Derive the radius of turning curve $R = V^2/(125\,f)$ and solve a numerical.
- Explain the system of runway and taxiway lighting with a neat sketch.

Quick revision: Runway width 18/23/30/45 m | effective gradient ≤ 1 % | transverse 1.5–2 % | RESA 90 m | Strip 60 m beyond | RET 30°, $R \geq 550$ m | $f = 0.13$.